

Playstation 3 Architecture and Development Tools

Powered by
The Cell Processor

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Agenda

- Introduction
- The multi-core era
- Playstation3 Architecture
- Cell Broadband Engine Processor
 - Cell Architecture
 - How games are using SPUs
 - Cell SDK
- RSX Graphics Processor
 - PSGL
 - Cg
- COLLADA
- Playstation Edge

Introduction

- **Developing games for consoles**
 - **Restrict to professional certificated developers**
 - **Development kits are expensive**
 - Nintendo Wii ~US\$ 2.000,00
 - Playstation 3 ~ US\$ 30.000,00
 - **Development kits are necessary**
 - Development kits contains software and hardware
 - You need the hardware to deploy and test your games

Introduction

- **In this lecture we will focus on**
 - **The SDKs, APIs and Tools used by professional developers to create games for the Playstation 3**
- **But almost all the SDKs, APIs and Tools used on the Playstation 3 are based on open standards**
 - **Cell Processor, OpenGL ES, Cg, COLLADA**
 - **Everything is also available to you!**

The multi-core era

- **Microprocessors are approaching the physical limits of semiconductors**
 - Small gains in processor performance from frequency scaling
- **One possible solution**
 - Increase the number of cores
- **We are in the multi-core era!!!**
 - Intel Core2 Duo, AMD X2, IBM Cell
 - Quad cores are coming
 - Single core processors are vanishing

Multi-core on consoles

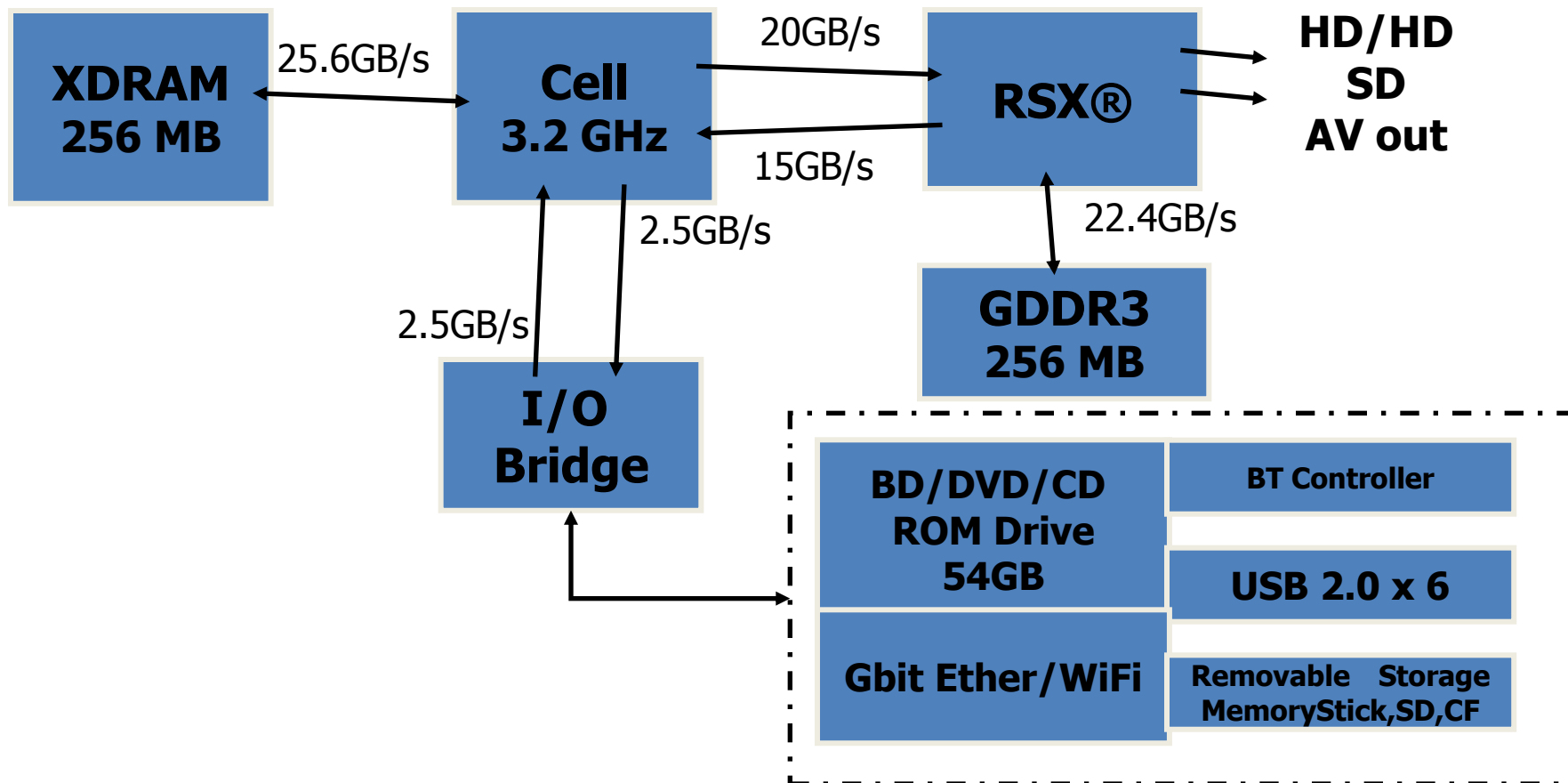
- **Playstation 3**
 - 9 cores (Cell Processor)
- **Xbox 360**
 - 3 cores (PowerPC based)
- **In the next generation all consoles should be multi-core!!!**

Playstation 3 Architecture

- **CPU: Cell Processor**
 - PowerPC-base Core @3.2GHz
 - 6 x accessible SPEs @3.2GHz
 - 1 SPE runs in a special mode (OS)
 - 1 of 8 SPEs disabled to improve production yields
- **GPU: RSX @550MHz (based on GeForce 7 series)**
 - Full HD (up to 1080p) x 2 channels
 - Multi-way programmable parallel floating point shader pipelines
- **Memory:**
 - 256MB XDR Main RAM @3.2GHz
 - 256MB GDDR3 VRAM @700MHz
- **System Floating Point Performance 2 TFLOPS**
- **Sound: Dolby 5.1ch, DTS, LPCM, etc**
- **Communications: Ethernet, Wi-Fi, Bluetooth**
- **Storage: Deattachable HDD slot**
- **Disc Media: CD/DVD/Blu-ray**



Playstation3 Architecture



The Cell Processor

Cell Broadband Engine Processor

- The CBE(Cell Broadband Engine) processor is the result of a collaboration between Sony, Toshiba and IBM
 - Alliance formed in 2000 and design center opened in 2001
 - First implementation in 2004
 - Investments approaching US\$400 million



TOSHIBA



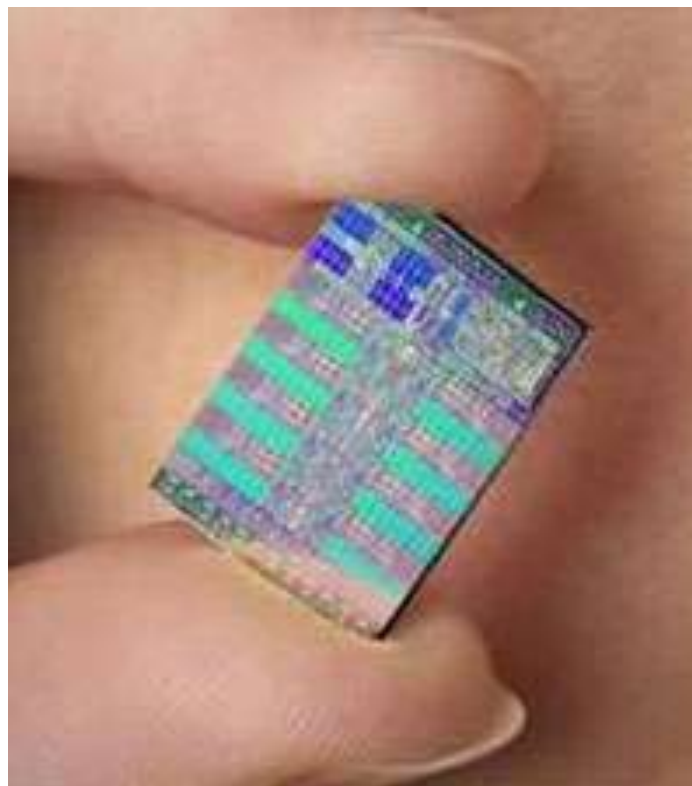
Cell Broadband Engine Processor

- **Heterogeneous single-chip multiprocessor**
 - Nine processor elements operating on a shared, coherent memory
- **Designed to support a very broad range of applications**
- **Overcomes three important limitations of contemporary microprocessors**
 - Power use, memory use and clock frequency

Cell Broadband Engine Processor

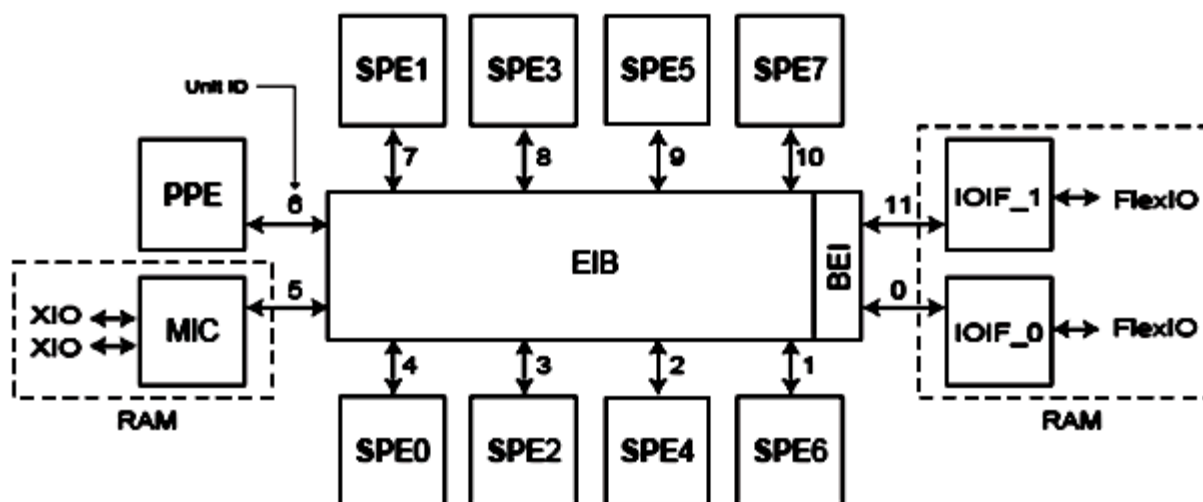
- **Power use**
 - Non Homogenous Coherent Multiprocessor
 - Improve power efficiency at approximately the same rate as the performance increase
- **Memory usage**
 - Asynchronous DMA transfers
 - 3-level SPE memory structure (main storage, local stores, and large register files)
- **Clock Frequency**
 - Specialize the PPE for control-intensive tasks and the SPEs for compute-intensive tasks
 - Run at high frequencies without excessive overhead

Cell Broadband Engine Processor



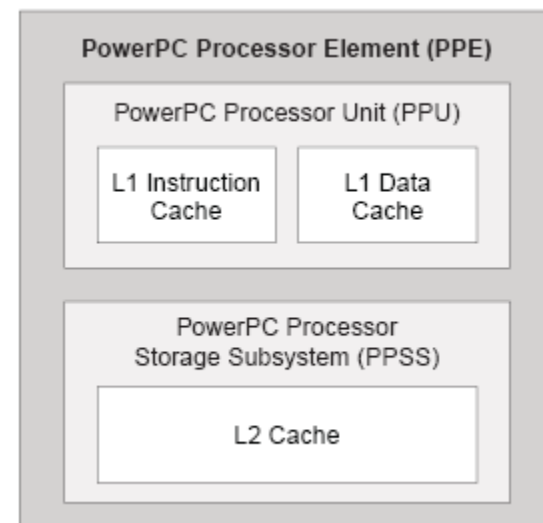
Cell Architecture

- **Heterogeneous single-chip multiprocessor**
 - 1x PPE (PowerPC Processor Element)
 - 8x SPE (Synergistic Processor Element)
- “It’s not a collection of different processors, but a synergistic whole”, Michael Perrone, IBM



Cell Architecture

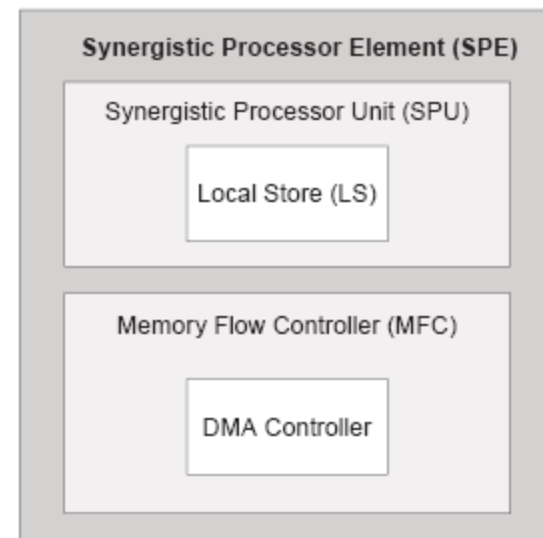
- **PPE (PowerPC Processor Element)**
 - **64-bit PowerPC Architecture RISC core**
 - General purpose processor
 - **Dual Thread**
 - Two way multi-processor with shared dataflow
 - **32 x 128 bit registers**
 - **2x 32KB L1 Caches (Instruction/Data)**
 - **512KB L2 Cache (Instruction and data)**
 - **VMX (Vector/SIMD multimedia extensions)**



Cell Architecture

● SPE (Synergic Processor Element)

- **128-bit RISC core**
 - Execute a new SIMD instruction set
 - Specialized for data-rich compute intensive SIMD and scalar applications
- **128 x 128 bit registers**
- **256KB Local Store (Instruction/Data)**
 - Coherent with main storage
 - SPU can only access its local store

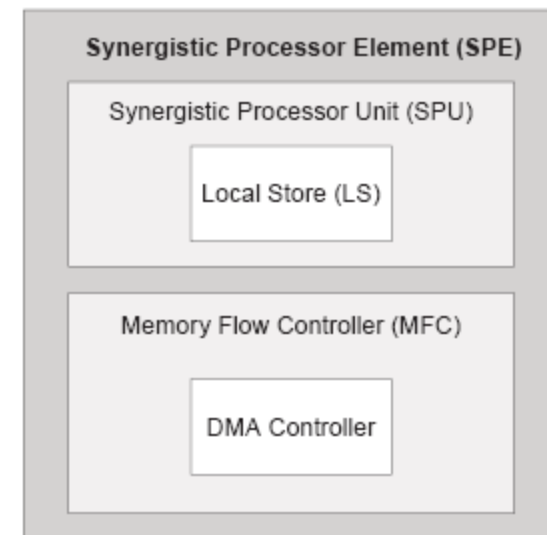


Cell Architecture

● SPE (Synergic Processor Element)

■ MFC

- DMA controller that moves instructions and data between its LS and main storage
- DMA 1/2/4/8/16 bytes up to 16KB
- Up to 16 in-flight DMA transfers

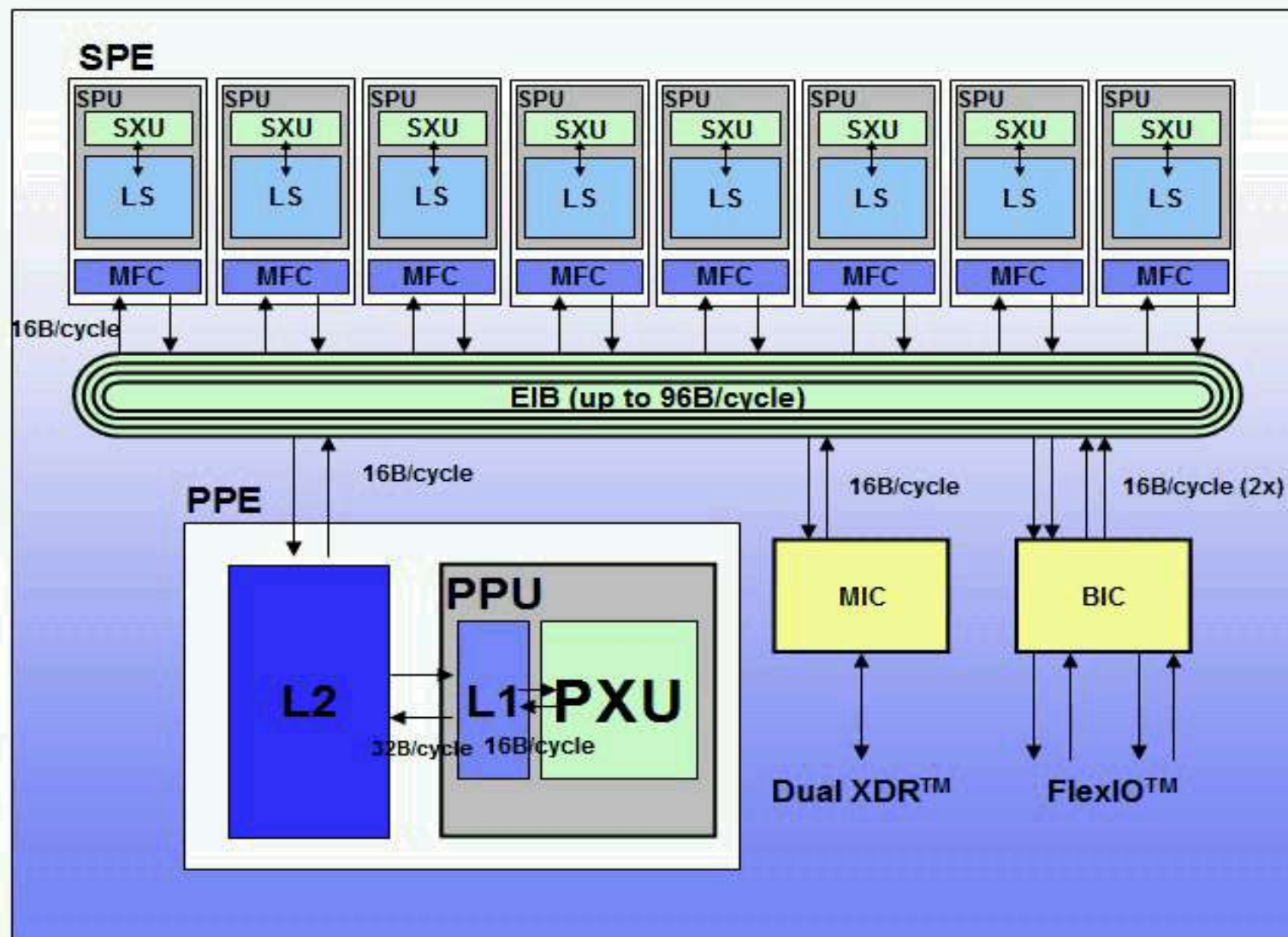


- The PS3 has 7 SPU's but only 6 are available to use

Cell Architecture

- **Element Interconnect Bus (EIB)**
 - Communication path for commands and data between all processors
 - Four 16-byte-wide data rings
- **Memory Interface Controller (MIC)**
 - Provides the interface between the EIB and the physical memory
- **Cell Broadband Engine Interface Unit (BEI)**
 - Provides a wide connection to external devices
 - Supports two Rambus FlexIO interfaces

Cell Architecture



Source: M. Gschwind et al., Hot Chips-17, August 2005

What makes Cell different?

- **Different programs running on the PPU and the SPU**
 - PPU: General purpose programs
 - SPU: Intensive computation programs
 - Both cooperating to carry out computations
- **SPE**
 - All the instructions are SIMD
 - SPU can only access its local store
 - Access to main memory done through asynchronous DMA

Big Fast Crowd [Reynolds, 2006]

Video

Simulating 12.000 boids at 60 fps

Big Fast Crowd [Reynolds, 2006]

- **Goal**
 - Simulate large groups of autonomous characters
- **Running on the Playstation 3**
 - Make use of the PPU, SPUs and RSX
 - All the simulation runs on the PPU and SPUs
 - Simulate up to 15.000 boids in real time
- **Individuals sorted by position into buckets**
 - Each SPU is used to update one bucket
 - SPUs are idle more than half of each frame!

How games are using the SPUUs?

MotorStorm Video

How games are using the SPUs?

- **MotorStorm SPU tasks**
 - Havok physics
 - Determination of object visibility
 - Concatenation of hierarchies
 - Billboard object culling and vertex buffer creation
 - Updating of particles and vertex buffer creation
 - Updating of vehicle dynamics
 - Audio (MultiStream)
 - Video decoding
- **Only uses 15%~20% of available SPU resources**

How games are using the SPUUs?

Lair Video

How games are using the SPUUs?

- **Lair SPU tasks**
 - **Physics**
 - **Skinning models**
 - **Culling triangles**
 - **Fluid Dynamics**
 - **Others**

How games are using the SPUs?

- **The SPUs are the key strenght of the PS3**
 - Ideal for offloading work from the PPU and RSX
 - Could be used to do a lot of different tasks
- **Many studios are trying to offload as much work as possible to the SPUs**
- **How to use the SPU?**
 - Direct create threads on the SPU and run your code
 - Run a kernel and a job manager on each SPU
 - Send jobs and tasks for each SPU
 - Sony has developed the SSW job manager for this purpose

Cell SDK

- **Complete Cell Broadband Engine development environment**
 - Documentation, libraries, samples, tools, IDE and a full system simulator for PC
 - Compatible with Fedora Core distribution
- **You don't need a Cell processor to program for the IBM Cell**

Cell SDK

○ Documentation

- Programming Hand Book
- SPE Runtime Management Library
- PPU & SPU Language Extension
- Tutorials

○ Libraries

- SPE Runtime management Library
- SPE Libraries: FFT, gmath, matrix, surface, sync, vector

○ Samples

- Many SPU samples
- Optimizing code on SPU samples (Euler)

Cell SDK

Tools

- IBM XL C/C++ Compiler
- GNU based C/C++ compiler
- GNU GDB
- GNU based binutils (assembler, linker, others)

IDE

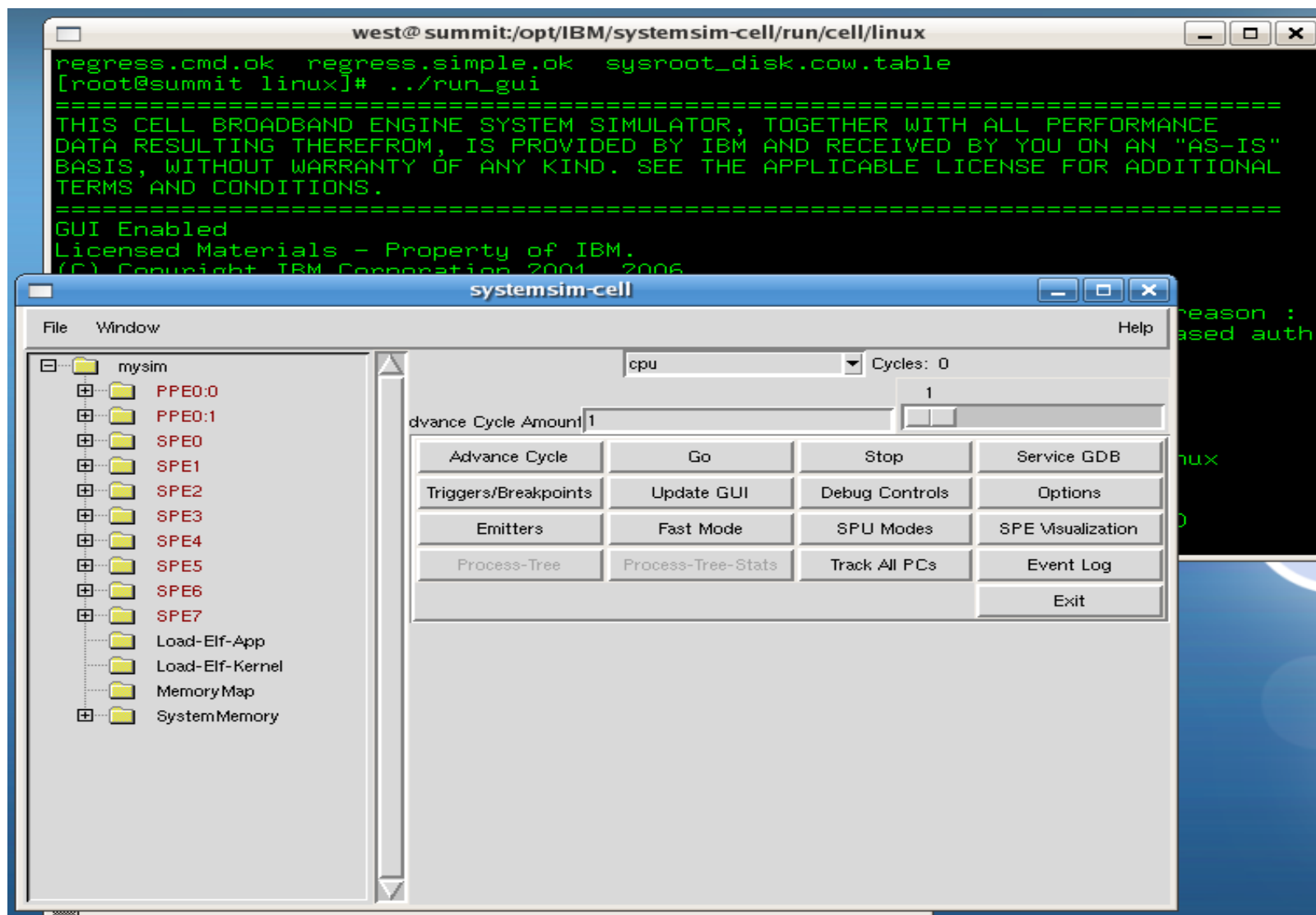
- Eclipse 3.1.1
- CDT (C/C++) Plugin
- IBM Cell System Simulator Plugin

Cell SDK

● System Simulator

- Full system simulator (emulates the behavior of a Cell Processor)
- Provides modes of functional-only and performance simulation
 - Fast Mode/Simple Mode/Pipeline Mode

Cell SDK



Linux on the PS3

- Since 2000 Sony is promoting Linux on the PS2
- There are some distributions available for the PS3
 - Fedora
 - Yellow Dog
 - Ubuntu
 - Gentoo

The RSX® Graphics Processor



- **Based on nVidia G70 architecture**
 - **@550 MHz**
 - **Fully programmable pipeline**
 - Supports shader model 3.0
 - Independent pixel/vertex shader architecture
 - Multi-way programmable parallel floating-point shader pipelines
 - **256MB GDDR3 dedicated video memory @650 MHz**
- **High Definition**
 - **720p/1080p**
- **Sony implemented a hypervisor to restrict RSX access on Linux =(**

PSGL

- **High-level graphics library for PlayStation3**
- **Based on OpenGL ES 1.0**
 - **Officially passed ES 1.0 conformance test**
 - **OpenGL ES 2.0 was not ready yet**
- **Add programmable pipeline to OpenGL ES 1.0**

PSGL

- **Why OpenGL ES?**
 - **Embrace an industry standard**
 - **Excellent specifications**
 - **Well-defined behavior**
 - **Industry collaboration**
 - **Conformance tests for quality**
 - **Expertise available**

PSGL

- **Supports many extensions**
 - **OpenGL ES 1.1 extensions**
 - **Programmable pipeline with Cg**
 - **Primitive/rendering extensions**
 - **Instancing, Primitive Restart, Queries, Conditional Rendering**
 - **Texture extensions**
 - **Floating Point, DXT, 3D, Non Power of 2, Anisotropic, Depth, Vertex Textures**
 - **Synchronization extensions**
 - **Synchronize with the PPU, SPU or another GPU**
 - **Fences, Events**
 - **Others...**

Cg: C for Graphics (nVidia)

- **High-level shading language created by nVIDIA**
 - Very similar to the Microsoft's HLSL
- **RSX supports Cg 1.5**
 - Has a specific compiler for the PS3
- **Great tools for developers**
 - FX Composer 2.0
 - nVidia Shader Perf



The COLLADA Format

COLLADA

- **No file format covered all the Next-Gen features**
 - **Multiple texture sets and values per vertex**
 - **Polygons, triangles, tri strips and fans**
 - **Curves (Splines)**
 - **Animation, skinning, blending, morphing**
 - **Shaders, effects**
 - **Physics**
- **COLLADA was designed to solve this**

COLLADA

- **Intermediate Digital Asset Exchange format**
 - Defines an open standard XML schema for exchanging digital assets
- **COLLADA is an industry standard**
 - Originally created by Sony Computer Entertainment
 - Adopted as industry standard by The Khronos Group
- **COLLADA 1.4.1 specification released on June 2006**
 - 298 pages (English/Japanese)
- **Supported by many DCC Tools**
 - 3D Studio Max, Maya, Softimage XSI, Blender

Why not a binary format?

● Binary files

- Must be specific optimized for the target Platform/API
- Difficult to debug
- Expensive to create

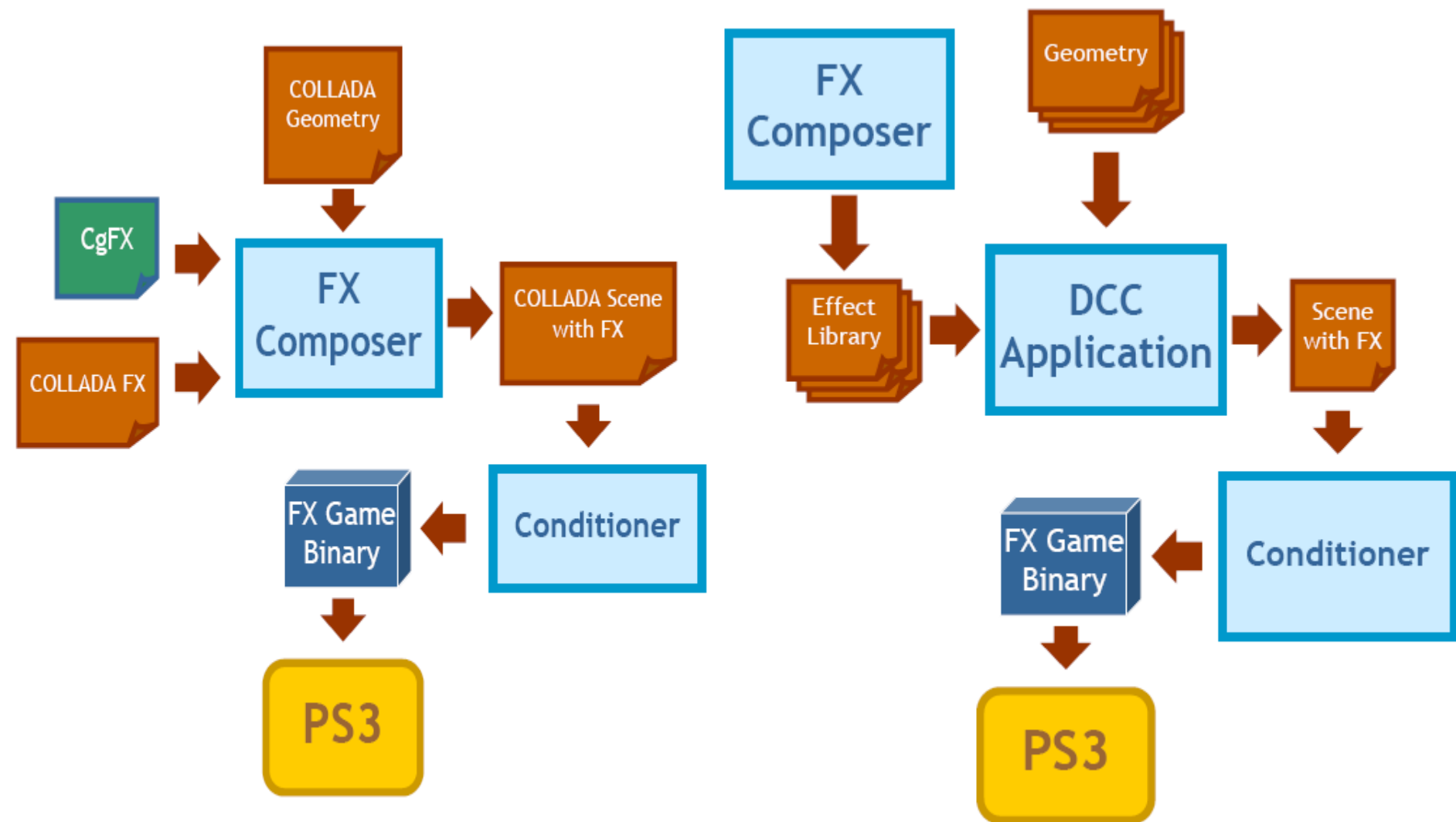
● XML files

- Very easy to debug / Human readable
- Can use schemas to validate the models
- Changes in the format are easy to handle
- Don't need to worry about optimizations
- Binary files can be generated targeting specific platforms

.DAE Cube Example

```
<library type="GEOMETRY">
  <geometry name="box">
    <mesh>
      <source id="box-Pos">
        <array id="box-Position-array" type="float" count="24">
          -0.5 0.5 0.5 ... (vertex data)
        </array>
      <technique profile="COMMON">
        <accessor source="#box-Position-array" count="8" stride="3">
          <param name="X" type="float" />
          <param name="Y" type="float" />
          <param name="Z" type="float" />
        </accessor>
      </technique>
    </source>
    <polygons> ... </polygons>
  </mesh>
</geometry>
</library>
```

COLLADA



GDC 2006, Khronos Group

GDC 2006, Khronos Group

COLLADA

● COLLADA FX

- First cross-platform standard shader and effects definition written in XML
- Next generation lighting, shading and texturing
- High level effects and shaders
- Support for all shader models

● COLLADA Physics

- Enables data interchange between Ageia (PhysX), Havok, Bullet, ODE and others
- Rigid Body, Dynamics Rag Dolls, Constraints, Collision Volumes

Who are using COLLADA ?



Autodesk



SOFTIMAGE



Overview of Playstation3 Development Tools

Playstation3 SDK

- Different from previous Playstations SDKs, the PS3 SDK uses many open standards
 - Cell SDK
 - PSGL (Playstation Graphics Library)
 - Cg (C for graphics)
 - COLLADA
- Only available to professional certificated developers

Playstation 3 Edge

- **New development tools for the Playstation 3**
 - “First party tech teams will be transferring technology to the general playstation 3 development public”, Mark Cerny
- **SPU Systems**
 - Animation engine (Many SPU systems)
 - Geometry system
 - Skinning
 - Triange culling
 - Blend shapes
 - Data compression (ZLib based)
- **GCM replay**
 - Powerful RSX analysis, debugging and profiling tool
 - Allows speculative performance analysis

Questions?

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"For what is a man profited, if he shall gain the whole world, and lose his own soul? or what shall a man give in exchange for his soul?" Matthew 16:26